Parth Garg

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Skills

Languages: Python, Java, C, C++, JavaScript, SQL
Web: React, Next.js, Node.js, HTML, CSS, Flutter, FastAPI
Data & ML: NumPy, Pandas, Matplotlib, PyTorch, scikit-learn, XGBoost, LightGBM, OpenCV, LangChain, RAG, LLMs
Database & Cloud: PostgreSQL, Supabase, SQLite, ChromaDB (vector DB), Firebase, AWS (EC2, S3, IAM)
Tools: Git, Linux, Docker, Kubernetes, CI/CD, VS Code, Streamlit, Ollama
Concepts: Data Structures & Algorithms, OOP, System Design, Full-Stack Development, Testing, Deep Learning, NLP

Education

Vellore Institute of Technology, Vellore **2024 – 2028**
B.Tech in Computer Science and Engineering (Information Security) *CGPA: 9.12 / 10*

Experience

BAMKO **May 2026 – Jun 2026**
Summer Intern *Dehradun, India*
- Built an LLM voice bot and full-stack speech features for a Quality Management System used by iLeads, using Sarvam speech-to-text and Groq for the language model step.
- Built the backend: a BullMQ and Redis job queue for STT, a 25-table PostgreSQL schema (Prisma) and REST APIs, deployed with Docker on AWS EC2.
- Fixed production bugs across four live systems (QMS, HR management, voice bot, call-center dialer) and contributed to system design for the HR management system.

Fintoric Capital **May 2025 – Jun 2025**
Data Analyst Intern *Delhi, India*
- Backtested three options strategies (Iron Condor, B120 and ORB) against historical market data in Pandas.
- Ran the firm's custom in-house models through the same backtests to check whether they would hold up going forward.
- Plotted daily and cumulative returns in Matplotlib so the team could compare strategies over the same periods.

Projects

Adaptive Block Routing, Hybrid Transformer-Mamba Model *Python, PyTorch, Mamba, CUDA* *GitHub*
- Built a hybrid language model in PyTorch where a Gumbel-Softmax router with entropy-controlled temperature sends each block of tokens to either attention or a Mamba state space layer (Mixture-of-Experts style routing).
- Introduced Cross-Block State Continuity, which carries Mamba's recurrent state across routing boundaries. WikiText-103 validation perplexity fell 21.7% (219.01 to 171.60), averaged over 3 seeds.
- 2.7x the throughput of a Transformer-only baseline at 1024 tokens; 16M-parameter model trained on one RTX 2050.

TunePilot, Autonomous LLM Fine-Tuning Orchestrator *Python, Hugging Face, QLoRA, SQLite, Kaggle GPUs* *GitHub*
- Built a terminal agent that picks a base model from Hugging Face, checks whether it fits in VRAM, writes the QLoRA or SFT config and launches training on free Kaggle dual T4 GPUs.
- Tracked every job in a SQLite state machine with retries and backoff, scanned notebooks for secrets before upload, and showed expected GPU hours and memory for approval before any run started.

Morph, Environment Fault Isolation Engine *Python, FastAPI, React, TypeScript, Linux cgroups, tc/netem* *GitHub*
- Built an engine that works out which environment difference (OS, network latency, locale, CPU or memory limits) makes a test fail, re-running it under controlled conditions with Linux cgroups, tc/netem and a TCP proxy, with anytime-valid sequential tests that hold false positives at alpha.
- Wrote a probabilistic bisection search with a Bayesian dose-response model that finds failure thresholds (for example, a lock lease that breaks above 134 ms of latency) with a 4 ms median error, against 21 ms for plain bisection over 40 runs.

Publications & Patents

An Efficient Handcrafted Feature-Driven Ensemble Learning Approach for Retinal Disease Screening
First author, International Conference on Next Generation Electronics (NELEX) IEEE 2026

A System for Real Time Environmental Perception and Assistance for a Visually Impaired User
Patent application, published 2026, App. No. 202641010249 *Indian Patent Office*

Achievements & Certifications

Winner, AI Solutions Track, Code2Create Hackathon **ACM-VIT | 2025**

Top 15 in Zodiak CTF and Best Write-up award for Re:Play **z0diak | 2026**

Supervised Machine Learning: Regression and Classification, Andrew Ng **Coursera | 2025**

Leadership & Extracurricular

TAM Club, VIT **2025 – Present**
Research Head *Vellore, India*
- Point of contact for the club's Agentic AI workshop, attended by 250+ participants.
- Designed and built the website for Code Cortex, a 30-hour ML hackathon (300+ participants), with zero failures.
- Work on the club's research papers, from ideation and experiments to technical writing.